Talagrand's discrete-convexity conjecture

Rewritten for readability from the OpenAI manuscript of the same title

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Abstract

We prove Talagrand's discrete-convexity conjecture. There is a universal integer k (namely $k = 2^{75}$) such that, whenever an arbitrary family $\mathcal{D}$ of subsets of $\{1, \dots, N\}$ has Bernoulli product measure $\mu_p(\mathcal{D}) \geq 1 - 1/k$, the sets not contained in a union of k members of $\mathcal{D}$ admit a cover of total cost at most $1/2$ at the same density p.

1 The result and the shape of its proof

1.1 Setting and statement

Let $N \geq 1$ and $[N] = \{1, \dots, N\}$. We identify a subset of $[N]$ with its indicator vector in $\{0,1\}^N$ whenever convenient. For $p \in (0,1)$, μ_p is the probability measure on $2^{[N]}$ under which each coordinate is present independently with probability p; that is, $\mu_p(\{S\}) = p^{|S|}(1-p)^{N-|S|}$.

Definition 1.1. Let $\mathcal{D}, \mathcal{A}, \mathcal{G} \subseteq 2^{[N]}$, let $k \geq 1$ be an integer, and let $\rho \in (0,1)$.

(a) The *exceptional family* is

$$E_k(\mathcal{D}) = \{S \subseteq [N] : S \nsubseteq D_1 \cup \cdots \cup D_k \text{ for all } D_1, \dots, D_k \in \mathcal{D}\}. \tag{1.1}$$

The members $D_1, \dots, D_k$ may repeat and need not be disjoint.

(b) $\mathcal{G}$ *covers* $\mathcal{A}$ if every $S \in \mathcal{A}$ contains some $I \in \mathcal{G}$. The members of $\mathcal{G}$ are called *generators*.

(c) The *cost* of $\mathcal{G}$ at density ρ is $c_\rho(\mathcal{G}) = \sum_{I \in \mathcal{G}} \rho^{|I|}$.

(d) $\mathcal{A}$ is *ρ-small* if it has a cover $\mathcal{G}$ with $c_\rho(\mathcal{G}) \leq 1/2$.

The same words are used with a finite ground set X in place of $[N]$.

Three immediate facts are used repeatedly.

(O1) *Subadditivity.* $c_\rho(\mathcal{G} \cup \mathcal{G}') \leq c_\rho(\mathcal{G}) + c_\rho(\mathcal{G}')$. If $\mathcal{G}$ covers $\mathcal{A}$ and $\mathcal{G}'$ covers $\mathcal{A}'$, then $\mathcal{G} \cup \mathcal{G}'$ covers $\mathcal{A} \cup \mathcal{A}'$. A cover of $\mathcal{A}$ covers every subfamily of $\mathcal{A}$. The empty family covers the empty family at cost 0.

(O2) *Monotonicity in the density.* If $\rho \leq \rho'$ then $c_\rho(\mathcal{G}) \leq c_{\rho'}(\mathcal{G})$. Hence ρ'-small implies ρ-small: smallness at a *larger* density is the *stronger* property.

(O3) *Smallness bounds the measure.* Since $\mu_p\{S : I \subseteq S\} = p^{|I|}$, a cover $\mathcal{G}$ of $\mathcal{A}$ gives $\mu_p(\mathcal{A}) \leq \sum_{I \in \mathcal{G}} p^{|I|} = c_p(\mathcal{G})$ by a union bound. So a p-small family has μ_p-measure at most $1/2$, with an explicit witness.

Theorem 1.2 (Main theorem)**.** *Set $k = 2^{75}$. For every integer $N \geq 1$, every $p \in (0,1)$, and every family $\mathcal{D} \subseteq 2^{[N]}$,*

$$\mu_p(\mathcal{D}) \geq 1 - \frac{1}{k} \quad \Longrightarrow \quad E_k(\mathcal{D}) \text{ is } p\text{-small.}$$

The constant k is independent of N, p, and $\mathcal{D}$. The family $\mathcal{D}$ is arbitrary; no monotonicity is assumed.

Example. If $\mathcal{D} = \{D : |D| \leq d\}$, then $E_k(\mathcal{D}) = \{S : |S| > kd\}$: a set lies in a union of k sets of size at most d exactly when its size is at most kd. The theorem asserts that, once $|D| \leq d$ has probability at least $1 - 1/k$, the event $|S| > kd$ is not only unlikely but has a cheap family of containment witnesses.

1.2 Proof at a glance

The proof has two layers and one combinatorial tool.

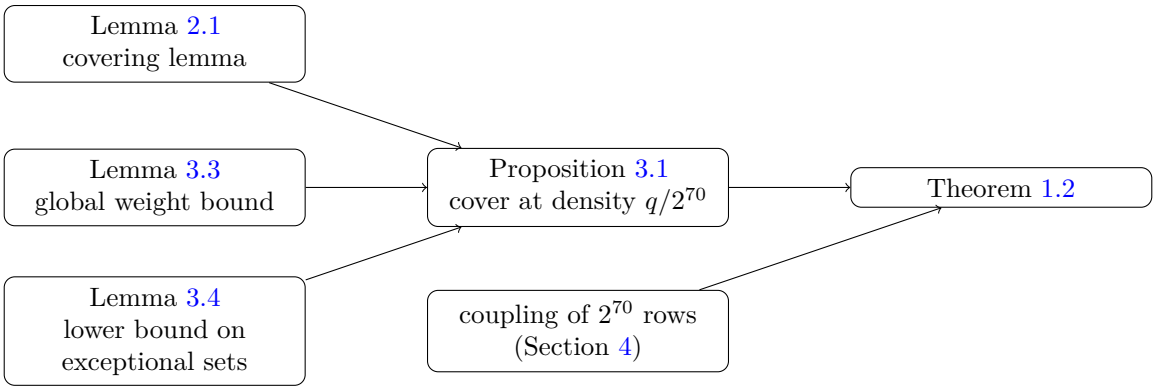
Outer layer: recovering the density (Section 4). Proposition 3.1 is a version of the theorem that loses a constant factor $L = 2^{70}$ in the density: if $\mu_q(\mathcal{F}) \geq 1/2$, then $E_{32}(\mathcal{F})$ is (q/L)-small. To get back to density p we apply it at the larger density $q = Lp$. We build, on one probability space, L random sets $R_1, \ldots, R_L$ that each have law μ_p but are pairwise disjoint, so that their union has law μ_{Lp}. Let $\mathcal{F}$ be the family of all unions of L members of $\mathcal{D}$. A union bound over the L rows gives $\mu_{Lp}(\mathcal{F}) \geq 1 - L/k = 31/32$, and $E_{32}(\mathcal{F}) = E_{32L}(\mathcal{D}) = E_k(\mathcal{D})$. The proposition then gives smallness at density $Lp/L = p$. When $Lp \geq 1$, a second coupling shows directly that $E_k(\mathcal{D}) = \varnothing$.

Inner layer: a cover at density q/L (Section 3). Write $f = \mathbf{1}_{\mathcal{F}}$. To each $U \subseteq [N]$ we attach a weight $w(U) = b(U)^2 \geq 0$, a rescaled squared biased Fourier coefficient of f. Two facts drive the argument.

- *Global bound* (Lemma 3.3, by Parseval): $\sum_U q^{|U|} w(U) \leq 1$.
- *Lower bound on exceptional sets* (Lemma 3.4, by a signed product measure on $32 \times N$ arrays): every $S \in E_{32}(\mathcal{F})$ satisfies $\sum_{\varnothing \neq U \subseteq S} w(U)^{16} \geq 2^{-32}$.

The cover is built as follows. Sets U with $w(U) \geq 2^{-64|U|}$ are taken as generators; the global bound makes this cheap. The remaining sets are sorted into bins $\mathcal{H}_{r,h}$ by size r and dyadic weight $w(U) \approx 2^{-h}$, where necessarily $h \geq 64r$. The global bound limits the size of each bin, and the covering lemma gives a cheap cover of all sets containing at least 2^{12h} members of the bin. A set S containing no generator contains fewer than 2^{12h} members of each bin, so its sum of sixteenth powers is at most $\sum_{r,h} 2^{12h} \cdot 2^{-16h} < 2^{-32}$; by the lower bound, S is not exceptional.

The tool: a covering lemma (Section 2). Lemma 2.1 says: if a family $\mathcal{H}$ of r-sets has $|\mathcal{H}|\rho^r \leq 2^{h+1}$, then the sets containing at least 2^{12h} members of $\mathcal{H}$ have a cover of cost at most $3 \cdot 2^{-h}$. The cover is random and has three parts: small sets lying in unusually many edges; unions of randomly sampled pairs of edges; and unions of those long tuples of edges none of whose pairs was sampled. A long tuple has quadratically many pairs, so it is very unlikely to survive.



Sections 2, 3 and 4 are self-contained apart from Parseval's identity. Section 5 collects a consequence, the context, and a side result (Corollary 5.1) that is *not* used in the proof of Theorem 1.2 and that rests on cited results.

1.3 Notation changes relative to the source

To keep one meaning per symbol, a few letters of the source are renamed. In Lemma 2.1: the degree threshold is τ (source: T) and the length of a partial tuple is n (source: t). In Section 3: a μ_q-random point is ξ (source: X), a $32 \times N$ array is A (source: x), and the signed column mass is $\widetilde{P}$ (source: R). In Section 4: the conditional inclusion probability is θ (source: t). In Corollary 5.1: the intermediate density is p_1 (source: r). The mathematics is unchanged.

2 An elementary covering lemma

Lemma 2.1 (Covering lemma)**.** *Let X be a finite set, let $0 < \rho < 1$, and let r, h be integers with $1 \le r \le h$ and $h \ge 64$. Suppose that $\mathcal{H} \subseteq \binom{X}{r}$ (a family of r-element subsets of X, called* edges*) satisfies*

$$|\mathcal{H}|\rho^r \le 2^{h+1}.$$

Then the family

$$\mathcal{M} = \Big\{ S \subseteq X : |\{e \in \mathcal{H} : e \subseteq S\}| \ge 2^{12h} \Big\}$$

has a cover $\mathcal{G} \subseteq 2^X$ with $c_\rho(\mathcal{G}) \le 3 \cdot 2^{-h}$.

Where it is used. Once for each occupied bin in Step 3 of the proof of Proposition 3.1, with $X = [N]$, $\rho = q/2^{70}$ and $\mathcal{H} = \mathcal{H}_{r,h}$.

Plan. We build a random family $\mathcal{G} = \mathcal{C}_0 \cup \mathcal{C}_1 \cup \mathcal{C}_2$ that covers $\mathcal{M}$ for *every* outcome of the randomness, and show that its *expected* cost is at most $3 \cdot 2^{-h}$; some outcome then does at least as well. $\mathcal{C}_0$ is deterministic and its cost is bounded by double counting (Steps 1–2). Removing $\mathcal{C}_0$ leaves "regular" edges with a uniform extension bound (Step 3), which controls the costs of $\mathcal{C}_1$ (Step 4) and $\mathcal{C}_2$ (Steps 5–8).

Proof. Fix the parameters

$$m = 2^{12h}, \qquad \tau = 2^{4h}, \qquad \pi = 2^{-8h},$$

where π is a probability, not the circle constant. Note $\pi m = 2^{4h}$.

Step 1 (weighted degrees and a double count). For $I \subseteq X$ with $|I| \le r$, define the *weighted degree*

$$d(I) = |\{e \in \mathcal{H} : I \subseteq e\}|\, \rho^{r-|I|}.$$

Then

$$\sum_{\substack{I \subseteq X \\ |I| \le r}} d(I)\rho^{|I|} = \rho^r \sum_{\substack{I \subseteq X \\ |I| \le r}} |\{e \in \mathcal{H} : I \subseteq e\}| = \rho^r \sum_{e \in \mathcal{H}} |\{I : I \subseteq e\}| = 2^r |\mathcal{H}| \rho^r. \tag{2.1}$$

The middle equality exchanges the order of summation over pairs (I, e) with $I \subseteq e$; every subset of an edge has size at most r, and an edge has 2^r subsets.

Step 2 (high-degree generators $\mathcal{C}_0$). Let

$$\mathcal{C}_0 = \{I \subseteq X : |I| \le r,\ d(I) > \tau\}.$$

For $I \in \mathcal{C}_0$ we have $\rho^{|I|} < \tau^{-1} d(I)\rho^{|I|}$. Summing over $\mathcal{C}_0$, enlarging the sum to all I with $|I| \le r$ (the terms are nonnegative), and using (2.1), the hypothesis $|\mathcal{H}|\rho^r \le 2^{h+1}$, and $r \le h$:

$$c_\rho(\mathcal{C}_0) \le \tau^{-1} 2^r |\mathcal{H}| \rho^r \le 2^{-4h} \cdot 2^r \cdot 2^{h+1} = 2^{r+1-3h} \le 2^{1-2h}. \tag{2.2}$$

Step 3 (regular edges and the extension bound). Call an edge *regular* if it contains no member of $\mathcal{C}_0$, and let $\mathcal{H}' \subseteq \mathcal{H}$ be the family of regular edges. If $I \subseteq e$ for some $e \in \mathcal{H}'$, then $|I| \le r$ and $I \notin \mathcal{C}_0$, so

$$I \subseteq e \in \mathcal{H}' \quad \Longrightarrow \quad d(I) \le \tau. \tag{2.3}$$

Claim. For every $B \subseteq X$,

$$\sum_{e \in \mathcal{H}'} \rho^{|e \setminus B|} \le \tau \sum_{i=0}^{r} \binom{|B|}{i}. \tag{2.4}$$

Proof of the claim. Group the edges $e \in \mathcal{H}'$ according to the set $I = e \cap B$. If the group of I is nonempty, then $I \subseteq B$, $|I| \le r$, every edge e in the group has $|e \setminus B| = r - |I|$, and $d(I) \le \tau$ by (2.3). Hence the group contributes

$$\sum_{\substack{e \in \mathcal{H}' \\ e \cap B = I}} \rho^{|e \setminus B|} = |\{e \in \mathcal{H}' : e \cap B = I\}|\, \rho^{r-|I|} \le |\{e \in \mathcal{H} : I \subseteq e\}|\, \rho^{r-|I|} = d(I) \le \tau.$$

The number of groups is at most the number of subsets of B of size at most r, which is $\sum_{i=0}^{r} \binom{|B|}{i}$. This proves (2.4).

Step 4 (sampled pairs $\mathcal{C}_1$). For each ordered pair $(e, e') \in (\mathcal{H}')^2$, diagonal pairs included, let $\sigma(e, e') \in \{0, 1\}$ be independent random variables with $\mathbb{P}(\sigma(e, e') = 1) = \pi$. A pair with $\sigma(e, e') = 1$ is called *sampled*. Let

$$\mathcal{C}_1 = \{e \cup e' : (e, e') \in (\mathcal{H}')^2,\ \sigma(e, e') = 1\}.$$

Several pairs may have the same union, which is counted once in $c_\rho(\mathcal{C}_1)$; therefore $c_\rho(\mathcal{C}_1) \le \sum_{e,e'} \sigma(e, e')\rho^{|e \cup e'|}$. Applying (2.4) with $B = e$, so that $|B| = r$ and $\sum_{i=0}^{r} \binom{r}{i} = 2^r$, gives $\sum_{e' \in \mathcal{H}'} \rho^{|e' \setminus e|} \le 2^r \tau$. Since $|e \cup e'| = r + |e' \setminus e|$,

$$\begin{aligned} \mathbb{E}[c_\rho(\mathcal{C}_1)] &\le \pi \sum_{e \in \mathcal{H}'} \rho^r \sum_{e' \in \mathcal{H}'} \rho^{|e' \setminus e|} \le \pi\, |\mathcal{H}'| \rho^r\, 2^r \tau \\ &\le 2^{-8h} \cdot 2^{h+1} \cdot 2^r \cdot 2^{4h} = 2^{r+1-3h} \le 2^{1-2h}. \end{aligned} \tag{2.5}$$

Here we used $|\mathcal{H}'|\rho^r \le |\mathcal{H}|\rho^r \le 2^{h+1}$ and $r \le h$.

Step 5 (residual generators $\mathcal{C}_2$; the cover). Call an ordered tuple $(e_1, \dots, e_m)$ of *distinct* regular edges *unsampled* if $\sigma(e_i, e_j) = 0$ for all $1 \le i, j \le m$. Let

$$\mathcal{C}_2 = \{e_1 \cup \cdots \cup e_m : (e_1, \dots, e_m) \text{ an unsampled tuple of distinct regular edges}\}, \qquad \mathcal{G} = \mathcal{C}_0 \cup \mathcal{C}_1 \cup \mathcal{C}_2.$$

Claim: $\mathcal{G}$ covers $\mathcal{M}$ for every outcome of σ. Let $S \in \mathcal{M}$. If S contains a member of $\mathcal{C}_0$ we are done. Otherwise every edge $e \subseteq S$ is regular (a member of $\mathcal{C}_0$ inside e would lie inside S), so S contains at least m distinct regular edges; choose m of them, $e_1, \dots, e_m$. If some pair (e_i, e_j) is sampled, then $e_i \cup e_j \in \mathcal{C}_1$ and $e_i \cup e_j \subseteq S$. If no pair is sampled, then $e_1 \cup \cdots \cup e_m \in \mathcal{C}_2$ and it is contained in S.

Step 6 (total weight of tuples). For $0 \le n \le m$, let

$$W_n = \sum_{\substack{(e_1,\dots,e_n) \in (\mathcal{H}')^n \\ e_1,\dots,e_n \text{ distinct}}} \rho^{|e_1 \cup \cdots \cup e_n|}, \qquad W_0 = 1.$$

Claim: $W_{n+1} \le \tau(1 + rm)^r \, W_n$ for $0 \le n < m$, hence $W_m \le [\tau(1 + rm)^r]^m$. Fix a tuple $(e_1, \dots, e_n)$ of distinct regular edges with union B; then $|B| \le rn \le rm$. Each way of extending it by a further edge $e \in \mathcal{H}'$ has weight $\rho^{|B \cup e|} = \rho^{|B|}\rho^{|e \setminus B|}$. Summing over *all* $e \in \mathcal{H}'$ (dropping the requirement that e differ from $e_1, \dots, e_n$ only adds nonnegative terms) and using (2.4),

$$\sum_{e \in \mathcal{H}'} \rho^{|B \cup e|} \le \rho^{|B|} \, \tau \sum_{i=0}^{r} \binom{|B|}{i} \le \rho^{|B|} \, \tau (1 + |B|)^r \le \rho^{|B|} \, \tau (1 + rm)^r.$$

The middle inequality $\sum_{i=0}^{r} \binom{|B|}{i} \le (1+|B|)^r$ holds because the map sending a subset $\{x_1 < \cdots < x_i\}$ of B (with respect to a fixed order of B, $i \le r$) to the sequence $(x_1, \dots, x_i, *, \dots, *)$ of length r over the alphabet $B \cup \{*\}$ is injective. Summing over all tuples $(e_1, \dots, e_n)$ gives the claim, and iterating from $W_0 = 1$ gives the bound on W_m.

Step 7 (expected cost of $\mathcal{C}_2$). Fix a tuple $(e_1, \dots, e_m)$ of distinct regular edges. Because the edges are distinct, the m^2 ordered pairs (e_i, e_j) are m^2 *distinct* elements of $(\mathcal{H}')^2$, so the corresponding variables $\sigma(e_i, e_j)$ are independent and

$$\mathbb{P}\big((e_1, \dots, e_m) \text{ is unsampled}\big) = (1 - \pi)^{m^2} \le \exp(-\pi m^2) \le 2^{-\pi m^2},$$

using $1 - \pi \le \exp(-\pi)$ and $\exp(1) > 2$. Each member of $\mathcal{C}_2$ is the union of at least one unsampled tuple, so $c_\rho(\mathcal{C}_2)$ is at most the sum of $\rho^{|e_1 \cup \cdots \cup e_m|}$ over unsampled tuples. By linearity of expectation and Step 6,

$$\mathbb{E}[c_\rho(\mathcal{C}_2)] \le W_m \, (1 - \pi)^{m^2} \le [\tau(1 + rm)^r]^m \, 2^{-\pi m^2}.$$

Step 8 (numerics). We show the last bound is at most 2^{-m}. Taking $\log_2$ and dividing by m, it suffices that

$$\log_2 \tau + r \log_2(1 + rm) + 1 \le \pi m = 2^{4h}.$$

Using $r \le h$ and $h + 1 \le 2^h$,

$$1 + rm \le 1 + h \, 2^{12h} \le (h + 1) 2^{12h} \le 2^{13h},$$

so the left side is at most $4h + 13rh + 1 \le 4h + 13h^2 + 1 \le 18h^2$ (as $4h + 1 \le 5h^2$ for $h \ge 1$). Finally $h \le 2^h$ gives $18h^2 \le 18 \cdot 2^{2h} \le 2^{4h}$, the last step because $18 \le 2^{2h}$ (true for $h \ge 3$, in particular for $h \ge 64$). Therefore

$$\mathbb{E}[c_\rho(\mathcal{C}_2)] \le 2^{(4h + 13rh)m - \pi m^2} \le 2^{-m}. \tag{2.6}$$

Step 9 (conclusion). By subadditivity (O1) and (2.2), (2.5), (2.6),

$$\mathbb{E}[c_\rho(\mathcal{G})] \le 2^{1-2h} + 2^{1-2h} + 2^{-m} = 2^{2-2h} + 2^{-m} \le 2^{-h} + 2^{-h} \le 3 \cdot 2^{-h},$$

where $2^{2-2h} \le 2^{-h}$ because $h \ge 2$, and $2^{-m} \le 2^{-h}$ because $m = 2^{12h} \ge h$. The sample space $\{0,1\}^{(\mathcal{H}')^2}$ is finite, so some outcome of σ has $c_\rho(\mathcal{G})$ at most its expectation. By Step 5 that outcome gives a cover of $\mathcal{M}$. □

3 Weights for exceptional sets

This section proves the fixed-loss version of the main theorem.

Proposition 3.1. *Let $L = 2^{70}$. For every integer $N \ge 1$, every $q \in (0,1)$, and every $\mathcal{F} \subseteq 2^{[N]}$ satisfying $\mu_q(\mathcal{F}) \ge 1/2$, the family $E_{32}(\mathcal{F})$ is (q/L)-small.*

Where it is used. In Case 1 of the proof of Theorem 1.2, with $q = Lp$.

Throughout this section N, q and $\mathcal{F}$ are fixed, and $f = \mathbf{1}_{\mathcal{F}} \colon \{0,1\}^N \to \{0,1\}$.

Definition 3.2 (Weights)**.** For $U \subseteq [N]$, $z \in \{0,1\}^U$ and $y \in \{0,1\}^{[N]\setminus U}$, write $(z,y) \in \{0,1\}^N$ for the vector equal to z on U and to y off U. Let Y be a random vector with independent Bernoulli-q coordinates indexed by $[N] \setminus U$. Define

$$b(U) = (1-q)^{|U|} \sum_{z \in \{0,1\}^U} (-1)^{\sum_{i \in U} z_i}\, \mathbb{E}[f(z,Y)], \qquad w(U) = b(U)^2. \tag{3.1}$$

For $U = \varnothing$ the sum has one term and Y is a full μ_q-sample, so

$$b(\varnothing) = \mathbb{E}[f(Y)] = \mu_q(\mathcal{F}). \tag{3.2}$$

As a further illustration, $b(\{i\}) = (1-q)(\mathbb{E}[f(0,Y)] - \mathbb{E}[f(1,Y)])$.

3.1 The global weight bound

Lemma 3.3 (Global weight bound)**.** *For every $q \in (0,1)$ and every $\mathcal{F} \subseteq 2^{[N]}$,*

$$\sum_{U \subseteq [N]} q^{|U|} w(U) \le \mu_q(\mathcal{F}) \le 1. \tag{3.3}$$

Where it is used. Steps 1 and 2 of the proof of Proposition 3.1.

Plan. Identify $b(U)$ with a biased Fourier coefficient of f up to an explicit factor, then apply Parseval. The basis is the standard biased Fourier basis [7, Section 8.4]; its normalization is made explicit here.

Proof. Let $\xi = (\xi_1, \dots, \xi_N) \sim \mu_q$.

Step 1 (an orthonormal basis). For $i \in [N]$ put $g_i(x) = (q - x_i)/\sqrt{q(1-q)}$ and, for $U \subseteq [N]$,

$$\chi_U(x) = \prod_{i \in U} g_i(x) = \frac{\prod_{i \in U}(q - x_i)}{[q(1-q)]^{|U|/2}}.$$

Since $\mathbb{E}\xi_i = q$ and $\operatorname{Var}\xi_i = q(1-q)$, we have $\mathbb{E}[g_i(\xi)] = 0$ and $\mathbb{E}[g_i(\xi)^2] = 1$. For $U, V \subseteq [N]$, independence of the coordinates gives

$$\mathbb{E}[\chi_U(\xi)\chi_V(\xi)] = \prod_{i \in U \cap V} \mathbb{E}[g_i(\xi)^2] \prod_{i \in U \triangle V} \mathbb{E}[g_i(\xi)] = \begin{cases} 1, & U = V, \\ 0, & U \ne V. \end{cases}$$

Because $0 < q < 1$, μ_q has full support, so $\langle g, g' \rangle = \mathbb{E}[g(\xi)g'(\xi)]$ is an inner product on the 2^N-dimensional space of real functions on $\{0,1\}^N$. The 2^N functions χ_U are orthonormal, hence a basis. With $\widehat{f}(U) = \mathbb{E}[f(\xi)\chi_U(\xi)]$, Parseval's identity reads

$$\sum_{U \subseteq [N]} \widehat{f}(U)^2 = \mathbb{E}[f(\xi)^2]. \tag{3.4}$$

Step 2 (relation between $\widehat{f}$ and b). We claim

$$\widehat{f}(U) = \left(\frac{q}{1-q}\right)^{|U|/2} b(U). \tag{3.5}$$

For one Bernoulli-q bit and $z_i \in \{0,1\}$,

$$\mathbb{P}(\xi_i = z_i)\,(q - z_i) = q(1-q)(-1)^{z_i},$$

since for $z_i = 0$ the left side is $(1-q)q$ and for $z_i = 1$ it is $q(q-1)$. Conditioning on the coordinates of ξ in U (the remaining coordinates have the law of Y and are independent of them),

$$\begin{aligned}\widehat{f}(U) &= \sum_{z \in \{0,1\}^U} \frac{\prod_{i \in U} \mathbb{P}(\xi_i = z_i)(q - z_i)}{[q(1-q)]^{|U|/2}}\, \mathbb{E}[f(z, Y)] = [q(1-q)]^{|U|/2} \sum_{z \in \{0,1\}^U} (-1)^{\sum_{i \in U} z_i}\, \mathbb{E}[f(z,Y)] \\ &= [q(1-q)]^{|U|/2} (1-q)^{-|U|}\, b(U) = \left(\frac{q}{1-q}\right)^{|U|/2} b(U).\end{aligned}$$

Step 3 (conclusion). Squaring (3.5) gives $q^{|U|} w(U) = (1-q)^{|U|}\widehat{f}(U)^2 \le \widehat{f}(U)^2$. Summing over U and using (3.4) and $f^2 = f$,

$$\sum_{U \subseteq [N]} q^{|U|} w(U) \le \sum_{U \subseteq [N]} \widehat{f}(U)^2 = \mathbb{E}[f(\xi)^2] = \mu_q(\mathcal{F}) \le 1. \qquad \square$$

3.2 A lower bound on every exceptional set

Lemma 3.4 (Lower bound on exceptional sets)**.** *Assume $\mu_q(\mathcal{F}) \ge 1/2$. Then every $S \in E_{32}(\mathcal{F})$ satisfies*

$$\sum_{\varnothing \ne U \subseteq S} w(U)^{16} \ge 2^{-32}. \tag{3.6}$$

Where it is used. Step 5 of the proof of Proposition 3.1.

Plan. Put a signed product weight ν_S on $32 \times N$ arrays of bits so that (i) ν_S vanishes unless the 32 rows, read as sets, have union containing S, and (ii) ν_S is a sum of terms that factor across the rows. Property (i) and $S \in E_{32}(\mathcal{F})$ force $\sum_A \nu_S(A) \prod_j f(\text{row } j) = 0$; property (ii) evaluates this sum as $\sum_{U \subseteq S} (-1)^{|U|} b(U)^{32}$. Isolating $U = \varnothing$ gives the bound.

Proof. Fix $S \in E_{32}(\mathcal{F})$. An *array* is $A = (A_{ji}) \in \{0,1\}^{32 \times N}$; its j-th row is $A_j = (A_{j1}, \dots, A_{jN}) \in \{0,1\}^N$, and its i-th column is $A^{(i)} = (A_{1i}, \dots, A_{32,i}) \in \{0,1\}^{32}$.

Step 1 (column masses). For a column $y \in \{0,1\}^{32}$ let

$$P(y) = \prod_{j=1}^{32} q^{y_j}(1-q)^{1-y_j}, \qquad \widetilde{P}(y) = P(y) - (1-q)^{32}(-1)^{\sum_{j=1}^{32} y_j}.$$

P is the mass function of 32 independent Bernoulli-q bits, and $\widetilde{P}$ is a signed modification with

$$\widetilde{P}(0,\dots,0) = (1-q)^{32} - (1-q)^{32} = 0. \tag{3.7}$$

Define the signed weight of an array by

$$\nu_S(A) = \prod_{i\notin S} P(A^{(i)}) \prod_{i\in S} \widetilde{P}(A^{(i)}).$$

Step 2 (a vanishing sum). We claim

$$\sum_{A\in\{0,1\}^{32\times N}} \nu_S(A) \prod_{j=1}^{32} f(A_j) = 0. \tag{3.8}$$

Indeed, every term is zero. If $\nu_S(A) \neq 0$, then by (3.7) no column $A^{(i)}$ with $i \in S$ is all-zero, i.e. every $i \in S$ belongs to at least one of the sets $A_1, \dots, A_{32}$, so $S \subseteq A_1 \cup \cdots \cup A_{32}$. As $S \in E_{32}(\mathcal{F})$, the rows cannot all lie in $\mathcal{F}$, so $\prod_j f(A_j) = 0$. Only pointwise vanishing is used; the signs of ν_S play no role.

Step 3 (expanding the product over S). Expanding $\prod_{i\in S} \widetilde{P}(A^{(i)})$ and letting $U \subseteq S$ be the set of columns where the subtracted term is chosen,

$$\nu_S(A) = \sum_{U\subseteq S} (-1)^{|U|} \lambda_U(A), \qquad \lambda_U(A) = (1-q)^{32|U|} \prod_{i\in U} (-1)^{\sum_{j=1}^{32} A_{ji}} \prod_{i\notin U} P(A^{(i)}).$$

(Columns outside S and columns in $S \setminus U$ both carry P.)

Step 4 (factoring over rows). Regroup the three factors of $\lambda_U(A)$ by rows:

$$(1-q)^{32|U|} = \prod_{j=1}^{32} (1-q)^{|U|}, \quad \prod_{i\in U} (-1)^{\sum_j A_{ji}} = \prod_{j=1}^{32} (-1)^{\sum_{i\in U} A_{ji}}, \quad \prod_{i\notin U} P(A^{(i)}) = \prod_{j=1}^{32} \prod_{i\notin U} q^{A_{ji}} (1-q)^{1-A_{ji}}.$$

Hence $\lambda_U(A) = \prod_{j=1}^{32} \beta_U(A_j)$, where for $x \in \{0,1\}^N$

$$\beta_U(x) = (1-q)^{|U|} (-1)^{\sum_{i\in U} x_i} \prod_{i\notin U} q^{x_i} (1-q)^{1-x_i}.$$

Since the rows of A range independently over $\{0,1\}^N$,

$$\sum_A \lambda_U(A) \prod_{j=1}^{32} f(A_j) = \Big(\sum_{x\in\{0,1\}^N} \beta_U(x) f(x) \Big)^{32}.$$

Writing $x = (z, y)$ with $z \in \{0,1\}^U$ and $y \in \{0,1\}^{[N]\setminus U}$, the product over $i \notin U$ is the probability that $Y = y$, so

$$\sum_x \beta_U(x) f(x) = (1-q)^{|U|} \sum_{z\in\{0,1\}^U} (-1)^{\sum_{i\in U} z_i} \mathbb{E}[f(z,Y)] = b(U)$$

by (3.1). Combining with Step 3 and (3.8) (all sums are finite),

$$0 = \sum_{U\subseteq S} (-1)^{|U|} b(U)^{32}. \tag{3.9}$$

Step 5 (conclusion). The term $U = \varnothing$ in (3.9) is $b(\varnothing)^{32} = \mu_q(\mathcal{F})^{32} \geq 2^{-32}$ by (3.2) and the hypothesis. Moving the other terms to the other side and using the triangle inequality,

$$2^{-32} \leq b(\varnothing)^{32} = -\sum_{\varnothing\neq U\subseteq S} (-1)^{|U|} b(U)^{32} \leq \sum_{\varnothing\neq U\subseteq S} |b(U)|^{32} = \sum_{\varnothing\neq U\subseteq S} w(U)^{16}. \qquad \square$$

3.3 Proof of Proposition 3.1

Plan. Take heavy sets as generators (Step 1), sort the rest into bins (Step 2), cover each bin's "many members" family by Lemma 2.1 (Step 3), add up the costs (Step 4), and show via Lemma 3.4 that a set containing no generator is not exceptional (Step 5). The gap exploited is that fewer than 2^{12h} weights of size at most 2^{-h} contribute less than $2^{12h} \cdot 2^{-16h} = 2^{-4h}$ to a sum of sixteenth powers.

Proof of Proposition 3.1. Set $\rho = q/L = q\,2^{-70} \in (0,1)$.

Step 1 (heavy sets). Let

$$\mathcal{G}_0 = \{U \subseteq [N] : U \neq \varnothing,\ w(U) \geq 2^{-64|U|}\}.$$

For $U \in \mathcal{G}_0$ with $r = |U| \geq 1$,

$$\rho^r = q^r 2^{-70r} = 2^{-6r} q^r\, 2^{-64r} \leq 2^{-6r} q^r w(U) \leq 2^{-6} q^r w(U).$$

Summing over $U \in \mathcal{G}_0$ and applying Lemma 3.3,

$$c_\rho(\mathcal{G}_0) \leq 2^{-6} \sum_U q^{|U|} w(U) \leq 2^{-6}. \tag{3.10}$$

Step 2 (bins). For integers $r \geq 1$ and h, let

$$\mathcal{H}_{r,h} = \{U \subseteq [N] : |U| = r,\ U \notin \mathcal{G}_0,\ 2^{-(h+1)} < w(U) \leq 2^{-h}\} \subseteq \binom{[N]}{r}.$$

(i) Every nonempty $U \notin \mathcal{G}_0$ with $w(U) > 0$ lies in exactly one bin, since the intervals $(2^{-(h+1)}, 2^{-h}]$, $h \in \mathbb{Z}$, partition $(0, \infty)$.

(ii) Every occupied (nonempty) bin has $h \geq 64r$: a member U has $2^{-(h+1)} < w(U) < 2^{-64r}$ (strictly, because $U \notin \mathcal{G}_0$), so $h + 1 > 64r$, i.e. $h \geq 64r$ as these are integers.

(iii) Every occupied bin satisfies $|\mathcal{H}_{r,h}|\rho^r \leq 2^{h+1}$: by Lemma 3.3,

$$|\mathcal{H}_{r,h}|\, q^r 2^{-(h+1)} < \sum_{U \in \mathcal{H}_{r,h}} q^r w(U) \leq 1,$$

so $|\mathcal{H}_{r,h}| q^r < 2^{h+1}$, and $\rho^r \leq q^r$.

Step 3 (covering each bin). For an occupied bin we have $1 \leq r \leq h$ and $h \geq 64r \geq 64$ by (ii), and $|\mathcal{H}_{r,h}|\rho^r \leq 2^{h+1}$ by (iii). So Lemma 2.1 applies with $X = [N]$ and $\mathcal{H} = \mathcal{H}_{r,h}$ and gives a family $\mathcal{G}_{r,h}$ that covers

$$\{S \subseteq [N] : S \text{ contains at least } 2^{12h} \text{ members of } \mathcal{H}_{r,h}\}, \qquad c_\rho(\mathcal{G}_{r,h}) \leq 3 \cdot 2^{-h}.$$

Only finitely many bins are occupied, since $[N]$ has finitely many subsets. Put

$$\mathcal{G} = \mathcal{G}_0 \cup \bigcup_{\mathcal{H}_{r,h} \neq \varnothing} \mathcal{G}_{r,h}.$$

Step 4 (cost of $\mathcal{G}$). By (ii), the occupied bins are among the pairs (r,h) with $r \geq 1$, $h \geq 64r$, and

$$3 \sum_{r \geq 1} \sum_{h \geq 64r} 2^{-h} = 3 \sum_{r \geq 1} 2 \cdot 2^{-64r} = \frac{6 \cdot 2^{-64}}{1 - 2^{-64}} = \frac{6}{2^{64} - 1} < \frac{1}{4}. \tag{3.11}$$

By subadditivity (O1), (3.10) and (3.11),

$$c_\rho(\mathcal{G}) \le \frac{1}{64} + \frac{6}{2^{64}-1} < \frac{1}{2}.$$

Step 5 ($\mathcal{G}$ covers $E_{32}(\mathcal{F})$). We prove the contrapositive: let $S \subseteq [N]$ contain no member of $\mathcal{G}$; we show $S \notin E_{32}(\mathcal{F})$. Consider a nonempty $U \subseteq S$. Either $w(U) = 0$, or $U \in \mathcal{G}_0$, or by (i) U lies in exactly one occupied bin. The second case is excluded because S contains no member of $\mathcal{G}_0$. For each occupied bin, S contains fewer than 2^{12h} members of $\mathcal{H}_{r,h}$; otherwise, by Step 3, S would contain a member of $\mathcal{G}_{r,h} \subseteq \mathcal{G}$. Each member of $\mathcal{H}_{r,h}$ has $w(U)^{16} \le 2^{-16h}$. Therefore, by (ii),

$$\sum_{\varnothing \ne U \subseteq S} w(U)^{16} \le \sum_{r\ge 1} \sum_{h \ge 64r} 2^{12h}\, 2^{-16h} = \sum_{r\ge 1} \frac{2^{-256r}}{1-2^{-4}} = \frac{16}{15(2^{256}-1)} < 2^{-32}. \tag{3.12}$$

By Lemma 3.4 (whose hypothesis $\mu_q(\mathcal{F}) \ge 1/2$ holds), $S \notin E_{32}(\mathcal{F})$.

Steps 4 and 5 show that $\mathcal{G}$ is a cover of $E_{32}(\mathcal{F})$ of cost less than $1/2$ at density $\rho = q/L$. □

4 Returning to the original density

Plan. With $L = 2^{70}$, couple L random sets so that each has law μ_p while their union has a prescribed larger density. If $Lp < 1$ the union has law μ_{Lp} and Proposition 3.1 applies at $q = Lp$ to the family $\mathcal{F}$ of L-fold unions. If $Lp \ge 1$ the union can be made all of $[N]$, and the exceptional family is empty.

Proof of Theorem 1.2. Set $L = 2^{70}$, so $k = 32L = 2^{75}$, and suppose $\mu_p(\mathcal{D}) \ge 1 - 1/k$.

Step 0 (union bound over rows). Let $R_1, \dots, R_L$ be random subsets of $[N]$ on a common probability space, each with law μ_p; no independence between them is assumed. Then $\mathbb{P}(R_j \notin \mathcal{D}) \le 1/k$ for each j, so

$$\mathbb{P}(R_j \in \mathcal{D} \text{ for all } 1 \le j \le L) \ge 1 - \sum_{j=1}^{L} \mathbb{P}(R_j \notin \mathcal{D}) \ge 1 - \frac{L}{k} = \frac{31}{32}. \tag{4.1}$$

Case 1: $Lp < 1$.

Step 1.1 (the coupling). For each coordinate $i \in [N]$, independently across coordinates, choose a label $J_i \in \{0, 1, \dots, L\}$ with

$$\mathbb{P}(J_i = 0) = 1 - Lp, \qquad \mathbb{P}(J_i = j) = p \quad (1 \le j \le L).$$

These are nonnegative and sum to 1 because $Lp < 1$. Put $R_j = \{i : J_i = j\}$ for $1 \le j \le L$, and $R = R_1 \cup \cdots \cup R_L$. See Figure 1.

Step 1.2 (laws). Fix j. The indicators $\mathbf{1}[i \in R_j] = \mathbf{1}[J_i = j]$, $i \in [N]$, are functions of the independent labels J_i, hence independent, and each equals 1 with probability p. So R_j has law μ_p. Likewise $\mathbf{1}[i \in R] = \mathbf{1}[J_i \ne 0]$ are independent and equal 1 with probability Lp, so R has law μ_q with $q = Lp \in (0, 1)$.

Step 1.3 (the family of L-fold unions is likely at density q). Define

$$\mathcal{F} = \{D_1 \cup \cdots \cup D_L : D_1, \dots, D_L \in \mathcal{D}\}.$$

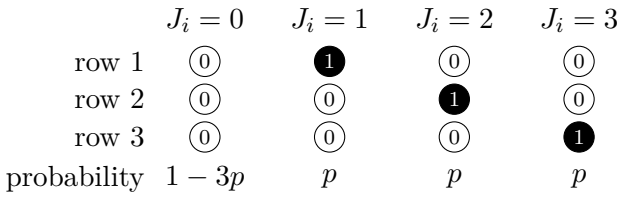


Figure 1: One coordinate of the sparse coupling, illustrated with three rows and $3p < 1$. Each row contains this coordinate with probability p; the union contains it with probability $3p$. Making these choices independently across coordinates gives the required product laws.

On the event in (4.1), $R = R_1 \cup \cdots \cup R_L$ belongs to $\mathcal{F}$. Since R has law μ_q,

$$\mu_q(\mathcal{F}) = \mathbb{P}(R \in \mathcal{F}) \geq \frac{31}{32} \geq \frac{1}{2}.$$

Step 1.4 (the exceptional families agree). A union of 32 members of $\mathcal{F}$ is a union of $32L$ members of $\mathcal{D}$; conversely, a union of $32L$ members of $\mathcal{D}$, grouped into 32 blocks of L, is a union of 32 members of $\mathcal{F}$. So the two collections of unions coincide, and by (1.1)

$$E_{32}(\mathcal{F}) = E_{32L}(\mathcal{D}) = E_k(\mathcal{D}).$$

Step 1.5 (conclusion of Case 1). By Step 1.3, Proposition 3.1 applies to $\mathcal{F}$ at density q: $E_{32}(\mathcal{F})$ is (q/L)-small. Since $q/L = p$ and $E_{32}(\mathcal{F}) = E_k(\mathcal{D})$, the family $E_k(\mathcal{D})$ is p-small.

Case 2: $Lp \geq 1$.

Step 2.1 (the coupling). Set

$$\theta = \frac{p - 1/L}{1 - 1/L}.$$

Then $\theta \in [0, 1)$: the numerator is nonnegative because $Lp \geq 1$, and smaller than the denominator because $p < 1$. At each coordinate i, independently across coordinates, choose a *mandatory row* uniformly among the L rows; put i into that row, and, conditionally on this choice, put i into each other row independently with probability θ. Let R_j be the set of coordinates put into row j.

Step 2.2 (laws and union). For each fixed j and i,

$$\mathbb{P}(i \in R_j) = \frac{1}{L} \cdot 1 + \left(1 - \frac{1}{L}\right)\theta = \frac{1}{L} + \left(p - \frac{1}{L}\right) = p.$$

The event $\{i \in R_j\}$ depends only on the choices made at coordinate i, and these are independent across coordinates; so R_j has law μ_p. Every coordinate lies in its mandatory row, so $R_1 \cup \cdots \cup R_L = [N]$ in every outcome.

Step 2.3 (conclusion of Case 2). By (4.1) the event that all L rows lie in $\mathcal{D}$ has positive probability, so some outcome gives $D_1, \ldots, D_L \in \mathcal{D}$ with $D_1 \cup \cdots \cup D_L = [N]$. Repeating this tuple 32 times gives $32L = k$ members of $\mathcal{D}$ with union $[N]$, which contains every $S \subseteq [N]$. Hence $E_k(\mathcal{D}) = \varnothing$, which the empty family covers at cost 0 (fact (O1)). The boundary case $Lp = 1$ is included, with $\theta = 0$. □

5 A consequence, context, and a side result

Nothing in this section is used in the proof of Theorem 1.2.

5.1 Positive selector processes

The connection to positive selector processes [11, Section 7] gives a concrete consequence of Theorem 1.2, recovering the positive-selector conclusion proved by Park and Pham [10, Theorem 1.2]. For a nonempty family $T \subseteq [0,\infty)^N$, define

$$\phi(S) = \sup_{t\in T} \sum_{i\in S} t_i, \qquad M = \mathbb{E}_{\mu_p}\phi,$$

and suppose $0 < M < \infty$. *Claim:* with $k = 2^{75}$, the family $\{S : \phi(S) \geq k^2 M\}$ is p-small.

Proof. Let $\mathcal{D} = \{D : \phi(D) < kM\}$.

(1) Markov's inequality gives $\mu_p\{\phi \geq kM\} \leq M/(kM) = 1/k$, so $\mu_p(\mathcal{D}) \geq 1 - 1/k$.

(2) Since all $t_i \geq 0$, ϕ is monotone ($S \subseteq S'$ implies $\phi(S) \leq \phi(S')$) and subadditive under unions ($\sum_{i\in A\cup B} t_i \leq \sum_{i\in A} t_i + \sum_{i\in B} t_i$ for each t, hence $\phi(A\cup B) \leq \phi(A) + \phi(B)$).

(3) If $S \subseteq D_1 \cup \cdots \cup D_k$ with all $D_j \in \mathcal{D}$, then by (2) $\phi(S) \leq \sum_j \phi(D_j) < k \cdot kM$. Hence $\{S : \phi(S) \geq k^2 M\} \subseteq E_k(\mathcal{D})$.

(4) By (1) and Theorem 1.2, $E_k(\mathcal{D})$ is p-small, and a cover of it covers the subfamily in (3) (fact (O1)). □

Thus the large-value event has explicit containment witnesses, beyond the small probability already supplied by Markov's inequality.

5.2 Context and related work

Origin of the problem. The name reflects the analogy with Talagrand's Gaussian convexity question [11, Conjecture 2.1]: can a bounded number of sums of a compact balanced set of sufficiently large Gaussian measure contain a convex set of Gaussian measure at least 1/2, uniformly in the dimension? (Balanced means closed under multiplication by scalars of absolute value at most one.) In the discrete problem, unions replace sums, and containment covers describe the exceptional sets. Talagrand formulated the discrete-convexity conjecture in [11, Conjecture 7.1]: a fixed number of members of a sufficiently likely family should cover every set outside a p-small exceptional family. The same original-density formulation appears as Research Problem 13.3.2 in [12], whose Proposition 13.3.3 explains its connection to positive selector processes. Theorem 1.2 proves this conjecture.

Related developments. Hua, Song, and Tudose resolve Talagrand's Gaussian convexity problem and derive a discrete conclusion with three unions and covering density p^C, for a universal constant C, when the starting family is sufficiently likely [5, Corollaries 1.3–1.4]. Ascoli, He, Park, and Talagrand give a threshold reformulation of discrete convexity and prove cases for several graph-containment properties [1, Proposition 1.5 and Theorems 1.10–1.13].

There is also a fractional version of smallness: a *fractional cover* of $\mathcal{A}$ of cost at most $1/2$ at density p is an assignment of nonnegative coefficients a_I with

$$\sum_{I \subseteq S} a_I \geq 1 \quad (S \in \mathcal{A}), \qquad \sum_{I \subseteq [N]} p^{|I|} a_I \leq \frac{1}{2}.$$

Talagrand introduced this relaxation alongside integral smallness [11, Sections 6 and 7]. Li resolves the fractional discrete-convexity problem through a quantitative obstruction to probability measures whose containment probabilities are bounded by $p^{|I|}$ [6, Theorem 1.9]. Combining fractional information with rounding, Park obtains an integral cover of $E_2(\mathcal{D})$ at density $p/(C \max\{1, \log\log(1/p)\})$ whenever $\mu_p(\mathcal{D}) > 1/2$, for a universal C [9, Corollary 3.5].

Antecedents of the method. With two rows instead of 32, the signed column mass $\widetilde{P}$ of Lemma 3.4 is Li's signed kernel after multiplication by the Bernoulli masses [6, Section 2]. Park also notes the extension of Li's argument to an even number of unions [9, Corollary 3.5, footnote 3]. What is new here is the combination of the higher powers obtained from 32 rows with Lemma 2.1, which yields a fixed density loss. The signed-operator approach to forbidden configurations and its biased Fourier basis have earlier forms in Friedgut [4, Section 2.2] and Ellis, Filmus, and Friedgut [2, Sections 2.2 and 4.1]. Unions of sampled edges also appear in the covering constructions of Fischer and Person [3, Sections 1.1 and 2]; Lemma 2.1 is proved in full above.

5.3 Two unions at constant density loss

Combining Li's result with the companion constant-factor rounding theorem gives the following two-union tradeoff. Compared with Theorem 1.2, it uses only two unions and a weaker measure hypothesis, but concludes smallness only at the lower density p/C (a weaker conclusion, by fact (O2)).

Corollary 5.1 (Two unions at constant density loss)**.** *There is a universal constant $C \geq 1$ such that, for every integer $N \geq 1$, every $p \in (0,1)$, and every family $\mathcal{D} \subseteq 2^{[N]}$,*

$$\mu_p(\mathcal{D}) > \frac{1}{2} \quad \Longrightarrow \quad E_2(\mathcal{D}) \text{ is } (p/C)\text{-small.}$$

Status of the proof. Unlike the rest of this paper, the proof is a chain of external results. The statements (i) and (iii) below are quoted exactly as the source uses them; this rewrite does not reprove them and has not compared them with the cited papers. Only the glue between them is checked here.

Proof. If $E_2(\mathcal{D}) = \varnothing$, the empty cover suffices. Otherwise:

(i) *$E_2(\mathcal{D})$ is a nonempty proper increasing family.* Since $\mu_p(\mathcal{D}) > 1/2$, $\mathcal{D} \neq \varnothing$, so $\varnothing \subseteq D \cup D$ for some $D \in \mathcal{D}$ and $\varnothing \notin E_2(\mathcal{D})$. If $S \in E_2(\mathcal{D})$ and $S \subseteq S'$, then S' is contained in no $D_1 \cup D_2$ either. By Li's result [6, Corollary 3.1] in Park's fractional-cover formulation [9, Theorem 3.4], this family has a fractional cover of cost at most $1/2$ at density $p_1 = p/2$.

(ii) *Capping.* Replace each coefficient a_I by $\min\{a_I, 1\}$. Coverage is preserved: if some $I \subseteq S$ had $a_I \geq 1$, the capped coefficient alone is 1; otherwise nothing changed for S. The cost does not increase.

(iii) *Rounding.* The companion expectation-threshold theorem [8, Theorem 1.1] then gives an integral cover of cost at most $1/2$ at density $p_1/(25 \cdot 512^4)$.

Since $p_1/(25 \cdot 512^4) = p/(50 \cdot 512^4)$, the constant $C = 50 \cdot 512^4$ suffices. □

Remark 5.2 (Consistency check for input (i); not part of the source's proof). The conclusion quoted in (i) can be confirmed with the tools of Section 3, taking $q = p$, $\mathcal{F} = \mathcal{D}$ and $\mu = \mu_p(\mathcal{D}) > 1/2$. The proof of Lemma 3.4 runs verbatim with 2 rows in place of 32 and gives $0 = \sum_{U \subseteq S} (-1)^{|U|} b(U)^2$ for $S \in E_2(\mathcal{D})$, hence $\sum_{\varnothing \neq U \subseteq S} w(U) \geq b(\varnothing)^2 = \mu^2$. By (3.5) and (3.4), $\sum_U (p/(1-p))^{|U|} w(U) = \mu$, so $\sum_{U \neq \varnothing} p^{|U|} w(U) \leq \mu - \mu^2$. Thus $a_U = w(U)/\mu^2$ $(U \neq \varnothing)$ is a fractional cover of $E_2(\mathcal{D})$ whose cost at density $p/2$ is at most $\frac{1}{2}(\mu - \mu^2)/\mu^2 = (1-\mu)/(2\mu) < 1/2$.

6 Verification checklist

No step of the chain leading to Theorem 1.2 is marked unverified. The steps that carry the weight, in the order a sceptical reader might check them:

1. **The signed identity** $0 = \sum_{U \subseteq S} (-1)^{|U|} b(U)^{32}$ (Lemma 3.4). Check that $\widetilde{P}$ vanishes on the all-zero column (3.7), that this forces every term of (3.8) to vanish (Step 2), and that each expanded term factors over the 32 rows with row factor exactly $b(U)$, including the split $(1-q)^{32|U|} = \prod_j (1-q)^{|U|}$ (Steps 3–4).

2. **The normalization** $\widehat{f}(U) = (q/(1-q))^{|U|/2} b(U)$, equation (3.5) (Lemma 3.3, Step 2). It is what turns Parseval into $\sum_U q^{|U|} w(U) \leq 1$ with the factor $q^{|U|}$, not a smaller one.

3. **The extension bound** (2.4) (Lemma 2.1, Step 3) and its two uses: with $B = e$ for the pair cost (2.5), and with B a prefix union for $W_{n+1} \leq \tau(1 + rm)^r W_n$ (Step 6).

4. **The survival probability and numerics** in Lemma 2.1, Steps 7–8: the m^2 pair indices of a tuple of distinct edges are distinct (so independence applies), and $4h + 13rh + 1 \leq 2^{4h} = \pi m$.

5. **The binning arithmetic** in the proof of Proposition 3.1: $h \geq 64r$ for occupied bins (Step 2(ii)), the hypothesis $|\mathcal{H}_{r,h}| \rho^r \leq 2^{h+1}$ of the covering lemma (Step 2(iii)), and the two geometric sums (3.11) and (3.12), in particular $\frac{16}{15(2^{256}-1)} < 2^{-32}$.

6. **The coupling** in the proof of Theorem 1.2: each row has law exactly μ_p and the union has law exactly μ_{Lp} (Step 1.2), respectively is $[N]$ (Step 2.2, including $\frac{1}{L} + (1 - \frac{1}{L})\theta = p$); the identity $E_{32}(\mathcal{F}) = E_k(\mathcal{D})$ (Step 1.4); and the constants $k = 32L$, $1 - L/k = 31/32$.

7. **External inputs.** The main chain uses only Parseval's identity. Corollary 5.1 alone depends on three cited results, quoted but not reproved here; see the status note before its proof.

References

[1] Ruben Ascoli, Xiaoyu He, Jinyoung Park, and Michel Talagrand, *A reformulation of the discrete Convexity Conjecture via k-thresholds*, arXiv:2608.11183v1, 2026. arXiv:2608.11183v1.

[2] David Ellis, Yuval Filmus, and Ehud Friedgut, *Triangle-intersecting families of graphs*, Journal of the European Mathematical Society **14** (2012), no. 3, 841–885. doi:10.4171/JEMS/320.

[3] Thomas Fischer and Yury Person, *Further remarks on fractional vs. expectation thresholds*, arXiv:2505.21782v1, 2025. doi:10.48550/arXiv.2505.21782.

[4] Ehud Friedgut, *On the measure of intersecting families, uniqueness and stability*, Combinatorica **28** (2008), no. 5, 503–528. doi:10.1007/s00493-008-2318-9.

[5] Dongming (Merrick) Hua, Antoine Song, and Stefan Tudose, *On Talagrand's Convexity Conjecture*, arXiv:2605.10908v2, 2026. arXiv:2605.10908v2.

[6] Chen Li, *On p-spread measures*, arXiv:2609.08967v1, 2026. arXiv:2609.08967v1.

[7] Ryan O'Donnell, *Analysis of Boolean Functions*, Cambridge University Press, Cambridge, 2014. Author's corrected text, 2021: arXiv:2105.10386v1.

[8] OpenAI, *Integral and fractional expectation thresholds are equivalent*, OpenAI Math Release preprint OAI:Integral-and-fractional-expectation-thresholds-are-equivalent-September-23-2026, 2026.

[9] Jinyoung Park, *A dimension-free comparison between expectation thresholds and fractional expectation thresholds*, arXiv:2609.14681v1, 2026. arXiv:2609.14681v1.

[10] Jinyoung Park and Huy Tuan Pham, *On a conjecture of Talagrand on selector processes and a consequence on positive empirical processes*, Annals of Mathematics **199** (2024), no. 3, 1293–1321. doi:10.4007/annals.2024.199.3.6.

[11] Michel Talagrand, *Are many small sets explicitly small?*, Proceedings of the Forty-Second ACM Symposium on Theory of Computing, ACM, New York, 2010, 13–36. doi:10.1145/1806689.1806693.

[12] Michel Talagrand, *Upper and Lower Bounds for Stochastic Processes: Decomposition Theorems*, 2nd ed., Ergebnisse der Mathematik und ihrer Grenzgebiete, 3. Folge, vol. 60, Springer, Cham, 2021. doi:10.1007/978-3-030-82595-9.